TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

BMIS2113 Information Technology

Infrastructure Assignment

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Programme & Group		RSD3 & G2
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Student	Part 1 (Bi-weekly)		Part II (Report)	Part III (Presentation)	Total (marks)
	10 + 20 (Group)	40 (Individual)	20 (Group)	10 (Individual)	100
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Comment:

1.0 Data Center Building Block

1.1 Existing Implementation and Impact

Existing Implementation	**Identified Issues**	**Impact on Business Operations**
All company databases and applications are housed in a single data center at Comet Retail Store's headquarters in Kota Kinabalu, Sabah.	The centralized architecture creates a Single Point of Failure (SPOF) since all business operations rely on a single physical location.	All ten retail locations and the online shopping platform will be affected simultaneously if the data center fails due to hardware malfunction, network disruption, or natural disasters, which may cause service interruptions and revenue loss.
A disaster recovery (DR) location is not mentioned in the current infrastructure.	In the event that the main data center fails, there is no backup location to carry on with business activities.	During a disaster, business continuity is impossible to sustain, resulting in protracted downtime and even data loss.
The organization is primarily reliant on internet purchasing, inventory management, CRM, and supplier collaboration.	Business-critical applications demand excellent system availability, yet redundancy is not emphasized.	Customers may encounter transaction difficulties, incorrect inventory information, delayed order processing, and negative shopping experiences.
No redundant power infrastructure, such as UPS, backup generators, or redundant Power Distribution Units (PDUs), is required.	Power outages could suddenly halt server operations.	Unexpected shutdowns can disrupt sales transactions, harm business data, and lower operational effectiveness.
No centralized or environmental monitoring systems are described.	IT administrators may not be able to spot faults immediately. Longer Mean Time To Repair (MTTR) raises downtime and maintenance expenses.	Longer Mean Time To Repair (MTTR) raises downtime and maintenance expenses.

1.2 Proposal and Justification

Proposed Solution	Justification (How It Resolves the Issue)	Alignment with Company Constraints, Malaysian Market Challenges, and Business Value
Establish a disaster recovery (DR) site in Kuching, Sarawak.	A secondary data center should constantly copy corporate data from Kota Kinabalu. During disasters or catastrophic hardware failures, corporate operations can be promptly restored from the disaster recovery site, maintaining business continuity and reducing downtime.	A DR site provides geographical redundancy between Sabah and Sarawak, lowering operating risks associated with localized calamities. It provides a practical and cost-effective disaster recovery solution for a mid-sized shop with ten locations.
Deploy redundant servers with automated failover clustering.	Configure application and database servers in clusters such that standby servers can take over if the primary server fails. This eliminates single points of failure (SPOFs) and improves service availability.	Automatic failover ensures that retail transactions, inventory synchronization, CRM, and online shopping services continue uninterrupted, increasing consumer happiness while lowering revenue loss due to hardware failure.
Implement Redundant Power Infrastructure.	Install twin Uninterruptible Power Supply (UPS) systems, backup diesel generators, dual power feeds, and redundant Power Distribution Units (PDUs) to ensure uninterrupted electrical power during utility outages.	Stable electrical power is critical for retail firms that rely on ongoing online and in-store transactions. This technology increases availability while safeguarding servers and databases from unplanned shutdowns.
Implement Centralized Infrastructure Monitoring. I	Install monitoring software to continuously monitor server performance, hardware health, storage use, network connectivity, temperature, humidity, and power conditions. Automated alerts allow IT managers to respond fast, reducing Mean Time To Repair (MTTR).	Early fault detection boosts operating efficiency, lowers maintenance costs, decreases service interruptions, and enables proactive infrastructure management.
Improve Data Center Physical Security	To prevent unwanted access to key infrastructure, use	Protecting corporate systems and customer information

	biometric access control, CCTV surveillance, electronic access tracking, secure server racks, and restricted access regulations.	enhances operational security, lowers physical security concerns, and boosts customer confidence in the company's digital offerings.

2.0 References

1. IBM. (2024). *What is disaster recovery?* IBM. https://www.ibm.com/think/topics/disaster-recovery

2. Microsoft. (2024). *Business continuity and disaster recovery*. Microsoft Learn. https://learn.microsoft.com/en-us/azure/architecture/framework/resiliency/overview

3. Microsoft. (2024). *Failover clustering overview*. Microsoft Learn. https://learn.microsoft.com/en-us/windows-server/failover-clustering/failover-clustering-overview

4. Schneider Electric. (2024). *Uninterruptible power supply (UPS)*. https://www.se.com/ww/en/work/products/explore/ups/